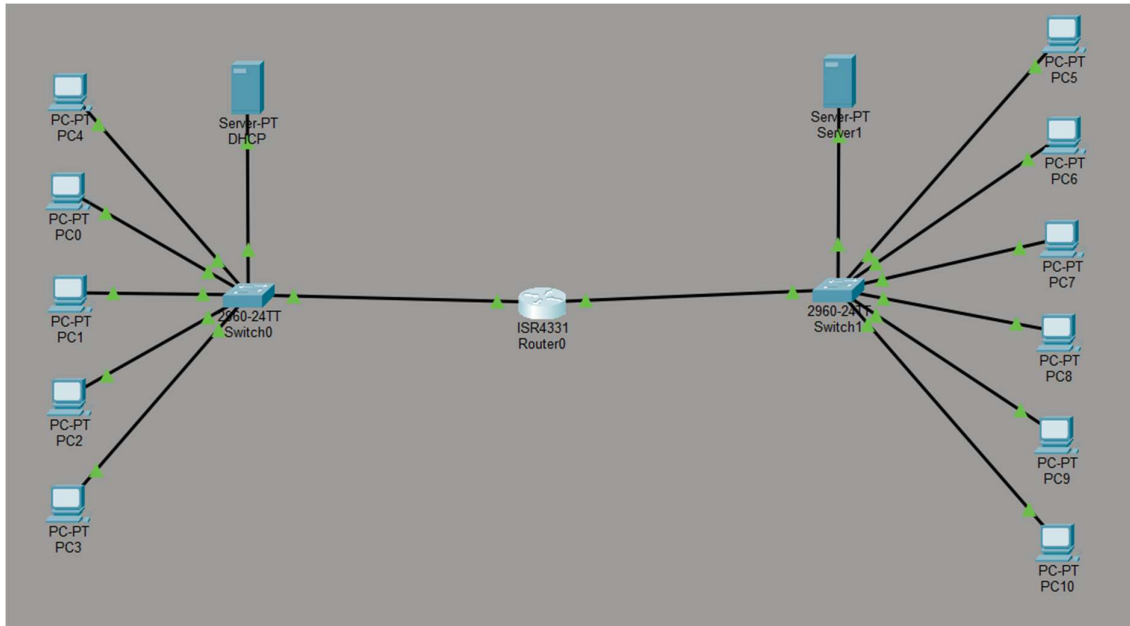


Experiment: 6

Objective: Design a network having two different LAN (LAN1 & LAN2).

LAN1 is having one 24 port switch along with one external DHCP Server. Also LAN1 is having 5 PCs with class B address. LAN2 is having one Switch and one DNS Server with 6 PCs with class C address. Configure the Router using CLI commands and Test the two different LAN using Ping command.



DHCP

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: serverPool

Default Gateway: 172.16.1.1

DNS Server: 192.168.2.2

Start IP Address: 172.16.1.3

Subnet Mask: 255.255.0.0

Maximum Number of Users: 253

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Buttons: Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
Lan2	192.168.2.1	192.168.2.2	192.168.2.20	255.255.255.0	236	0.0.0.0	0.0.0.0
serverPool	172.16.1.1	192.168.2.2	172.16.1.3	255.255.0.0	253	0.0.0.0	0.0.0.0

Server1

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS**
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

DNS

DNS Service ☒ On ☐ Off

Resource Records

Name Type

Address

No.	Name	Type	Detail
0	dns.com	A Record	192.168.2.2

DHCP

Physical Config Services **Desktop** Programming Attributes

IP Configuration X

IP Configuration

☐ DHCP ☒ Static

IPv4 Address

Subnet Mask

Default Gateway

DNS Server

DNS

Physical Config Services **Desktop** Programming Attributes

IP Configuration X

IP Configuration

☐ DHCP ☒ Static

IPv4 Address

Subnet Mask

Default Gateway

DNS Server

PC1

Physical Config **Desktop** Programming Attributes

Command Prompt X

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.22

Pinging 192.168.2.22 with 32 bytes of data:

Reply from 192.168.2.22: bytes=32 time<1ms TTL=127
Reply from 192.168.2.22: bytes=32 time<1ms TTL=127
Reply from 192.168.2.22: bytes=32 time<1ms TTL=127
Reply from 192.168.2.22: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.2.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
Router>en
Router#conf t
Router(config)#int g0/0/0
Router(config-if)#ip address 172.16.1.1 255.255.0.0
Router(config-if)#no sh
Router(config-if)#do wr
Router(config-if)#exit

Router(config)#int g0/0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no sh
Router(config-if)#do wr
Router(config-if)#ip helper-address 172.16.1.10
Router(config-if)#exit
```